CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building AI/LLM services and agentic workflows in Python and taking them to production on AWS with containerized, CI/CD-driven deployments. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by backend and cloud work across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Owned an LLM knowledge-extraction service end-to-end as sole developer, from data ingestion and chunking to prompt engineering and indexing, reporting directly to the CTO and working with SAP domain experts.
- Built agentic workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) on complex regulatory and technical documents, evaluated against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Replaced LLM calls with a deterministic, code-based pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service exposing the knowledge graph to AI agents with millisecond-latency retrieval, integrated into Claude Code.
- Developed Expert Interface, a web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, LLM APIs, MCP, FastAPI, Docker, Kubernetes, Azure, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes each request to vector database search, general LLM inference or summarization.
- Built the Python/FastAPI backend and a React front-end, tested the APIs with Postman, containerized the stack with Docker and deployed it to AWS with CDK (infrastructure as code); versioned in GitLab with CI/CD pipelines.
- Worked with business stakeholders to understand their decision-making workflows and translated their requirements into the solution design.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated testing of the PrismaChat RAG assistant's accuracy: generation and retrieval evaluation with RAGAS and PytrecEval against chemist-provided ground truth.
- Built serverless AWS infrastructure (Lambda, SQS, DynamoDB) with CDK to run the evaluations automatically.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, retrieval fine-tuning).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy on unseen data.

PROJECTS

Project: LLM Gateway | Python | FastAPI | LiteLLM | AWS | Terraform

- Provider-agnostic LLM API layer in Python/FastAPI over OpenAI, Anthropic and Google models via LiteLLM, with token budgets, retries and fallback between providers, and a LangGraph example workflow as client.
- Deployed to AWS (ECS Fargate) with Terraform; per-request latency, token and cost logging in CloudWatch with alarms, pytest and a small eval set in GitHub Actions CI/CD.

TECHNICAL SKILLSET

- **Programming & Backend:** Python, FastAPI, REST APIs, Python Microservices, SQL, Postman, React (beginner)
- **AI/LLM Services:** LLM APIs via LiteLLM, LangGraph Agentic Workflows, LangChain, MCP, RAG, GraphRAG, Prompt Engineering, Claude Code
- **Cloud & Infrastructure:** AWS (EC2, Lambda, SQS, DynamoDB, S3, VPC, CDK), Azure, Docker, Kubernetes, Infrastructure as Code (AWS CDK)
- **CI/CD & Quality:** Git, GitLab CI/CD, automated LLM evaluation (RAGAS, PytrecEval), API testing
- **Data & ML:** Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search), Pandas, PyTorch, Scikit-Learn, Hugging Face Transformers

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)